Supplementary Computational Certificate for the 8216 Three-Zone Bound

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Abstract

This supplement records the exact finite computation used in the Laurent–BKK stage of the accompanying paper. The committed JSON file contains every exponent vector, every Newton vertex, and the fifteen normalized Minkowski-sum volumes for each of three polynomial systems. A deterministic verifier reconstructs the data from the Laurent formulas and proves

$$\mathrm{MV}(\mathcal{P}_0) = 6808, \qquad \mathrm{MV}(\mathcal{P}_1) = 1568, \qquad \mathrm{MV}(\mathcal{P}_2) = 1248.$$

These values give $6808 + (1568 + 1248)/2 = 8216$.

1 Files and verification command

The certificate consists of

File	Contents
`verification/verify_certificate.py`	exact symbolic, support, hull, and volume verifier
`verification/mixed_volume_certificate.json`	stable machine-readable certificate

It is checked by

```
python3 verification/verify_certificate.py \
  --check verification/mixed_volume_certificate.json
```

The command regenerates the JSON and compares it byte for byte with the committed file. It reports

```
support/vertex counts: F=36/16, G=265/24, H2=1154/52,
H1_tilde=12360/54, P_epsilon=13/8, Hc_tilde=2692/74
normalized polarization sums: 163392, 37632, 29952
mixed volumes: 6808, 1568, 1248
regular positive three-zone bound: 8216
```

2 Exact Laurent support construction

All monomials have the form

$$z^a w^b R^c S^d, \qquad (a, b, c, d) \in \mathbb{Z}^4.$$

The verifier implements sparse Laurent arithmetic over an exact formal coefficient ring generated by

$$\alpha,\ \beta,\ p,\ c_-,\ \overline{c_-},\ c_+,\ \overline{c_+},\ \frac{\mu^2}{16B^2}.$$

The physical involution sends z, w, p to their inverses, exchanges each $c_\pm$ with its formal conjugate, and fixes the real parameters. Thus all coefficient cancellations are decided exactly, without numerical parameter substitution.

After substituting the three displayed flight numerators from the paper, sparse division in R verifies

$$(R-1)^3 \mid F_0, \qquad (R-1)^4 \nmid F_0, \qquad (R-1)^2 \mid G_0.$$

The quotients $F_L = F_0/(R-1)^3$ and $G_L = G_0/(R-1)^2$ generate the two basic supports. Contact supports are then obtained solely by union, integer Minkowski addition, derivative translation, and the determinant formulas in the paper. In particular, the covering-contact set is generated directly rather than replaced by a rectangular envelope.

Object	exponent vectors	Newton vertices
F_L	36	16
G_L	265	24
H_2	1154	52
$\widetilde{H}_1$	12360	54
P_ε	13	8
$\widetilde{H}_c$	2692	74

The JSON keys `supports` and `vertices` contain the corresponding complete lexicographically ordered lists.

3 Exact hull and volume checks

SciPy/Qhull is used only to propose hull vertices and a triangulation of each boundary. For every proposed nondegenerate facet, the verifier computes a primitive integer normal from cofactors and checks that all vertices lie in one closed supporting half-space. It chooses the rational barycenter of the vertices as an interior cone point. Every cone-simplex determinant is recomputed using `Fraction`; their absolute values sum to

$$\mathrm{Vol}_{\mathbb{Z}}(P) = 4!\,\mathrm{Vol}_{\mathrm{Eucl}}(P).$$

Degenerate zero-volume simplices introduced by a triangulation of a nonsimplicial rational facet are discarded exactly. An accepted normalized volume is required to be an integer.

For four polytopes $P_1, \dots, P_4$, the BKK convention in the paper is

$$\mathrm{MV}(P_1, P_2, P_3, P_4) = \frac{1}{4!} \sum_{\varnothing \neq I \subseteq \{1,2,3,4\}} (-1)^{4-|I|}\,\mathrm{Vol}_{\mathbb{Z}}\left(\sum_{i \in I} P_i\right). \tag{1}$$

The factor $1/4!$ converts normalized lattice volumes back to the coefficient convention of Bernstein's theorem.

4 The forty-five Minkowski-sum volumes

In each row below the subsets occur in mask order

$$1,\ 2,\ 12,\ 3,\ 13,\ 23,\ 123,\ 4,\ 14,\ 24,\ 124,\ 34,\ 134,\ 234,\ 1234.$$

The entries are normalized volumes $\mathrm{Vol}_{\mathbb{Z}}(P_I)$.

System	1	2	12	3	13
$\mathcal{P}_0$	192	2240	14784	14780	50364
$\mathcal{P}_1$	192	2240	14784	64	2528
$\mathcal{P}_2$	192	2240	14784	14780	50364

System	23	123	4	14	24
$\mathcal{P}_0$	104508	236540	207530	414450	640426
$\mathcal{P}_1$	12480	46496	39760	107264	200752
$\mathcal{P}_2$	104508	236540	64	2528	12480

System	124	34	134	234	1234
$\mathcal{P}_0$	1099938	1099878	1757342	2396278	3553822
$\mathcal{P}_1$	405504	95912	217912	379592	695672
$\mathcal{P}_2$	46496	46476	120620	225868	445612

Substitution into (1) gives

$$\begin{aligned} \mathcal{P}_0: &\quad 163392/24 = 6808, \\ \mathcal{P}_1: &\quad 37632/24 = 1568, \\ \mathcal{P}_2: &\quad 29952/24 = 1248. \end{aligned}$$

5 Certified implication

The two-step Khovanskii–Rolle inequality in the paper is

$$N^{+}_{\mathrm{reg},3z} \le \mathrm{MV}(\mathcal{P}_0) + \frac{1}{2}\big(\mathrm{MV}(\mathcal{P}_1) + \mathrm{MV}(\mathcal{P}_2)\big).$$

The exact values above therefore prove the finite computational implication

$$\boxed{N^{+}_{\mathrm{reg},3z} \le 6808 + \frac{1}{2}(1568 + 1248) = 8216}.$$